

Supplementary Material S1: Reproducibility Manifest, Data Models, and Verifier Issue Taxonomy

Antigravity Research Team

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This supplementary document accompanies the manuscript “**A Verifier-Gated Neuro-Symbolic Agent Creates Isomorphic Math Variants and Step Scaffolds**”. It provides full technical specifications, dataset provenance, random seeds, serialized data schemas, LLM execution configurations, verifier issue taxonomy, and post-hoc answer-audit protocols referenced in the main text.

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1 S1.1 Experiment Identifiers and Sampling Seeds

To ensure complete reproducibility of the empirical evaluation reported in Section 4, all experimental trials were executed against a fixed sample derived from deterministic pseudo-random shuffling.

- **Experiment Identifier:** `tri_dataset_200_seed_20260803`
- **Experiment Directory:** `experiments/tri_dataset_200_seed_20260803/`
- **Primary Sampling Seed:** 20260803
- **Dataset-Specific Shuffling Seeds:**
 - **GSM8K:** 20260803
 - **MATH Algebra:** 20260804
 - **PRM800K:** 20260805
- **Experimental Design:** 600 source items ($200 \text{ items/dataset} \times 3 \text{ datasets}$) $\times$ 2 generation modes (`isomorphic`, `scaffold`) $\times$ 2 LLM models (`gpt-4o-mini`, `deepseek-v3`) = 2,400 total experimental trials.

2 S1.2 Dataset Provenance and SHA-256 Manifest

Table 1 details the upstream source endpoints, local file paths, and cryptographic SHA-256 hashes for all evaluation datasets.

Table 1: Dataset provenance, local storage paths, and cryptographic SHA-256 manifests.

Dataset	Upstream Source	Split / Config	Local Path
GSM8K	Hugging Face <code>openai/gsm8k</code>	<code>main</code> , <code>test split</code>	<code>data/raw/gsm8k_test.jsonl</code>
MATH	Hugging Face EleutherAI/ <code>hendrycks_math</code>	<code>algebra</code> , <code>test split</code>	<code>data/raw/math_algebra_test.jsonl</code>
PRM800K	OpenAI <code>openai/prm800k</code>	<code>phase2_test</code>	<code>data/raw/prm800k_phase2_test.jsonl</code>

Direct Download Endpoints:

- **GSM8K:** https://raw.githubusercontent.com/openai/grade-school-math/master/grade_school_math/data/test.jsonl
- **MATH Algebra:** <https://people.eecs.berkeley.edu/~hendrycks/MATH.tar.gz>
- **PRM800K:** https://media.githubusercontent.com/media/openai/prm800k/main/prm800k/data/phase2_test.jsonl

3 S1.3 LLM Provider and Service Configurations

Table 2 summarizes the LLM service endpoints, sampling parameters, and context limits used during formalization and candidate generation.

Additional Runtime Execution Parameters:

Table 2: LLM service endpoints and hyperparameter configurations.

Pipeline Stage	Reported Model Identifier	Default Base URL	Temp.	Max Tokens
Formalization (Decomposer)	<code>gpt-4o-mini</code>	https://api.openai.com/v1	0.2	2,500
GPT Candidate Generation	<code>gpt-4o-mini</code>	https://api.openai.com/v1	0.2	1,800
DeepSeek Candidate Gen.	<code>deepseek-ai/DeepSeek-V3</code>	https://api.siliconflow.cn/v1	0.2	1,800

- **Concurrency Limit:** 4 parallel worker threads.
- **Maximum Agent Retries (k):** 2 retry attempts per trial (up to 3 total attempts including initial generation).
- **JSON Schema Repair Allowance:** Up to 2 automatic repair requests per API call if JSON parsing fails.
- **API Timeout:** 60 seconds per HTTP request.
- **Credential Isolation:** API keys are retrieved from environment variables (`OPENAI_API_KEY`, `SILICONFLOW_API_KEY`) or git-ignored `config.json` files, and are scrubbed from log files and public artifacts.

4 S1.4 Serialized Data Models and Controller States

The controller architecture formalizes problem items, steps, candidate responses, and verifier outputs using immutable data structures.

4.1 Controller Terminal States

The `MathAgent` controller outputs one of two terminal states:

1. **DELIVERED(candidate, verification_result, attempt_history):** Emitted when a generated candidate passes all SymPy verifier checks ($CVI \geq 0.60$, $SFS \geq 0.60$, real solvable roots, satisfied domain constraints).
2. **ABSTAINED(attempt_history, failure_reasons):** Emitted when all k attempts fail verification, triggering fail-safe abstention to prevent invalid items from reaching students.

4.2 Serialized Data Schema

- **Problem:** `{problem_id, text, variables, equations, steps, domain_constraints}`
- **Step:** `{step_id, name, description, equations, dependencies, concepts}`
- **Candidate:** `{question, variables, equations, answer, reasoning}`
- **VerificationResult:** `{accepted, solutions, cvi, isomorphism_score, issues}`
- **Attempt:** `{number, candidate, verification, reflection, error}`

5 S1.5 Verifier Failure Issue Taxonomy and Reflection Mappings

When candidate verification fails, the `SymbolicVerifier` categorizes the failure into an issue taxonomy code. The `ReflectionEngine` maps these codes into prompt constraints for the next attempt (Table 3).

Table 3: Verifier issue taxonomy codes and corresponding reflection prompt actions.

Issue Code	Failure Description	Reflection Prompt Action
<code>PARSE_ERROR</code>	Equation or JSON schema cannot be parsed	Enforce strict SymPy syntax and declared variables
<code>NO_SOLUTION_ERROR</code>	Equation system has no finite solution	Add numerical givens to uniquely determine solution
<code>SOLVER_ERROR</code>	SymPy internal solver exception or timeout	Simplify algebraic structure and eliminate circular dependencies
<code>NON_REAL_SOLUTION_ERROR</code>	Solutions contain complex imaginary parts	Ensure discriminant $\Delta \geq 0$ for real-valued roots
<code>DOMAIN_VIOLATION_ERROR</code>	Solutions violate active constraints ($x \leq 0$)	Regenerate coefficients so roots satisfy predicted domain
<code>UGLY_SOLUTIONS_ERROR</code>	Clean-Value Index ($CVI < 0.60$) is low	Restructure coefficients for small integers/fractions
<code>NON_ISOMORPHIC_ERROR</code>	Structural similarity ($SFS < 0.60$) is low	Preserve variable count, degree, and operator types
<code>TARGET_STEP_MISMATCH</code>	Scaffold targets wrong step ID	Align step equations strictly with <code>target_step</code>
<code>GENERATION_ERROR</code>	API network error or JSON failure	Re-prompt LLM with strict JSON schema instructions

6 S1.6 Post-Hoc Answer-Audit Implementation

To evaluate the mathematical consistency between the natural-language text generated by LLMs and the SymPy-verified solution, a post-hoc audit was implemented (`scripts/audit_paper_results.py`).

6.1 Audit Classification Categories

Each delivered candidate item was audited and assigned to one of three categories:

1. **Matched (Answer Agreement):** The candidate’s natural-language answer text (e.g., “ $x = 18$ ”) mechanically parses into a numeric value or tuple that matches the SymPy solver’s exact solution set.
2. **Confirmed Contradiction:** The parsed natural-language answer text explicitly contradicts every solution computed by SymPy (e.g., text states “ $x = 25$ ” but SymPy solver computed “ $x = 18$ ”).
3. **Mechanically Unverified:** The candidate’s text contains prose, complex units, or non-standard formatting that cannot be unambiguously parsed by regular expressions, retained in the denominator for conservative accuracy bounds.

6.2 Statistical Confidence Intervals

Proportions for contradiction rates and match rates are reported with **Wilson score 95% confidence intervals**:

$$CI_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (1)$$

where $z = 1.96$, $\hat{p}$ is the sample proportion, and n is the total delivered item count.

7 S1.7 Artifact Directory Map and Regeneration Commands

7.1 Artifact Directory Structure

Table 4: Artifact directory map for experimental outputs.

Relative Location	Description of Retained Contents
data/benchmark/	Fixed 200-item paper benchmark JSONL files (<code>gsm8k_200.jsonl</code> , etc.)
data/normalized/	Full normalized test splits for GSM8K, MATH Algebra, and PRM800K
web/	Standalone FastAPI server, HTML template, CSS styling, and JS visual
examples/demo_run.py	Dual-mode (Offline + Online API) Python demonstration script
tests/	Automated unit test suite (<code>test_agent.py</code> , <code>test_datasets.py</code>)
scripts/run_large_experiment.py	Benchmark execution pipeline for 2,400 experimental trials
scripts/analyze_experiment_for_paper.py	Statistical analysis, McNemar tests, CVI/SFS distribution tables
scripts/audit_paper_results.py	Post-hoc answer-consistency and evidence-chain audit script

7.2 Commands for Full Experiment Regeneration

To execute the full verification and analysis pipeline from a fresh installation:

```
# 1. Install package in editable mode
pip install -e .

# 2. Run unit test suite
python -m unittest discover -s tests -v

# 3. Run interactive Web UI server
python -m uvicorn web.server:app --host 0.0.0.0 --port 8000

# 4. Run statistical analysis and paper table generation
python scripts/analyze_experiment_for_paper.py

# 5. Run post-hoc answer consistency audit
python scripts/audit_paper_results.py
```